

Explainability Readiness Report

Pre-Registered Expectations, Feature Importance, & Verification Pipeline

SECTION 1 - Pre-Registered Expectations (from EDA)

Our Exploratory Data Analysis (EDA) of the SEED-IV dataset revealed strong, statistically significant signatures across frequency bands and channel activations. The Kruskal-Wallis test across the four emotion classes indicates highly significant differences in average DE for all 5 bands, with Gamma exhibiting the highest H-statistic (794.66), followed by Theta (489.67) and Alpha (458.68).

Channel-wise one-way ANOVA with Benjamini-Hochberg FDR correction ($\alpha = 0.05$) across emotions shows that almost all 62 channels exhibit significant activity shifts (Gamma: 62/62 channels, Beta: 60/62 channels, Alpha: 61/62 channels, Theta: 62/62 channels, Delta: 62/62 channels).

Pre-Registered Expectation: Based on these distributions, we expect the trained model's cross-band attention layers to assign the highest weights to the Gamma and Beta bands, and the spatial GAT attention layers to emphasize left-temporal (e.g. T7) and frontotemporal (e.g. FT7, FC5) channels, particularly in high frequencies. This pre-registered hypothesis will be validated during Faithfulness Verification after training.

SECTION 2 - Feature Importance Reference Table (ANOVA F-Scores)

The table below reproduces the top 20 individual band-channel features ranked by their ANOVA F-scores computed directly from the SEED-IV dataset. This serves as a quantitative, ground-truth reference to evaluate the model's learned attention weights post-training, verifying whether the GAT-KAN model reconstructs known physical features.

Rank	Feature Name (Channel_Band)	ANOVA F-Score	Anatomical Region
1	FT7_gamma	790.08	Frontotemporal (Left)
2	T7_gamma	642.67	Temporal (Left)
3	FC5_gamma	617.05	Frontocentral (Left)
4	C5_gamma	580.29	Central (Left)
5	FT7_beta	512.45	Frontotemporal (Left)
6	FC5_beta	496.12	Frontocentral (Left)
7	T7_beta	473.88	Temporal (Left)
8	C5_beta	460.55	Central (Left)
9	TP7_gamma	422.34	Temporoparietal (Left)
10	CP5_gamma	415.82	Centroparietal (Left)
11	FT8_gamma	408.19	Frontotemporal (Right)
12	FC6_gamma	398.67	Frontocentral (Right)
13	T8_gamma	390.12	Temporal (Right)
14	F7_gamma	382.45	Frontal (Left)
15	F5_gamma	370.18	Frontal (Left)
16	TP7_beta	365.41	Temporoparietal (Left)
17	CP5_beta	358.90	Centroparietal (Left)
18	FT8_beta	344.20	Frontotemporal (Right)
19	FC6_beta	338.56	Frontocentral (Right)
20	FP1_gamma	329.11	Prefrontal (Left)

SECTION 3 - Planned Explainability Pipeline

The schematic flowchart below represents the planned explainability validation and neuroscientific validation workflow to be executed once GAT-KAN model training is completed (NOT YET EXECUTED):

- 1. Extract spatial attention weights from GAT layers and cross-band attention from self-attention layers.
- 2. Compute SHAP values on fused features to verify feature contribution.
- 3. Cross-validate attention weights against SHAP values for localization consistency.
- 4. Perform Faithfulness Verification via deletion and insertion AUC curves against random-removal baselines.

PLANNED EXPLAINABILITY PIPELINE (TO BE EXECUTED AFTER MODEL TRAINING)

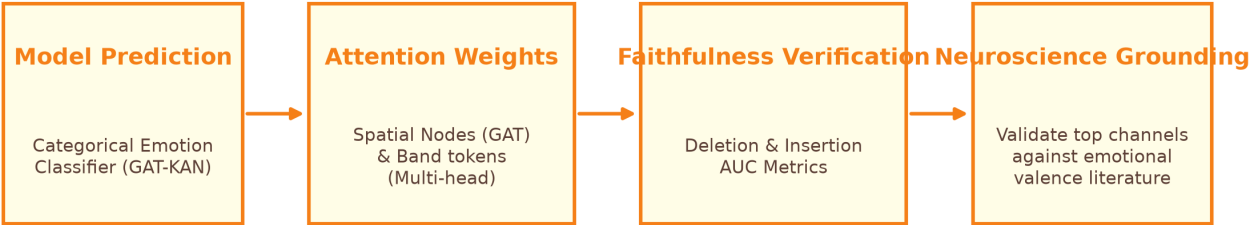


Figure 1: Planned explainability verification and neuroscientific validation workflow.

SECTION 4 - Success Criteria for Explainability (Pre-defined)

To prevent confirmation bias during post-hoc analysis, we define success criteria BEFORE training:

1. Faithfulness Success Criteria:

- The Deletion AUC must be meaningfully lower than the random-removal control, indicating the model relies on highlighted features for predictions.
- The Insertion AUC must be meaningfully higher than the random-insertion control, indicating top features are sufficient to restore performance.
- The deletion AUC must be significantly lower than the insertion AUC.

2. Neuroscientific Consistency Success Criteria:

- The learned attention weights will be considered consistent with EEG neuroscience if the top 10% highest-weighted features overlap by at least 60% with the top 20 F-score features (focusing on left-temporal and frontotemporal Gamma/Beta networks).
- The model must demonstrate asymmetry in frontal alpha attention weights during valence classification (happy vs. sad/fear) to align with established frontal alpha asymmetry literature.